

Definition - The Tachometer use for measuring the rotational speed or angular velocity of the machine which is coupled to it. It works on the principle of relative motion between the magnetic field and shaft of the coupled device.

The relative motion induces the EMF in the coil which is placed between the constant magnetic field of the permanent magnet. The developed EMF is directly proportional to the speed of the shaft.

"Mechanical and electrical are the two types of the Tachometer. The mechanical Tachometer measures the speed of shaft regarding revolution per minutes."

"The electrical Tachometer converts the angular velocity into an electrical voltage. The electrical Tachometer has more advantages over the mechanical Tachometer."

Tachometer can be classified into two types

- ① AC Tachometer
- ② DC Tachometer

Photoelectric Tachometer

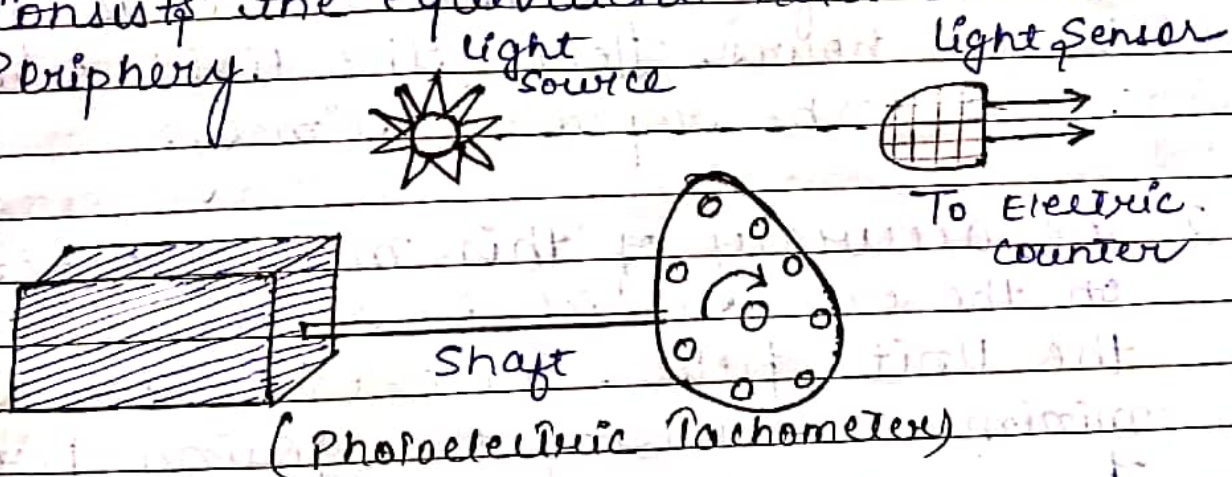
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The Tachometer which uses the light for measuring the speed of rotation of shaft or disc of machines is known as Photoelectric Tachometer.

The opaque disc with holes on its periphery, light source and laser are the essential parts of the Photoelectric Tachometer.

The Tachometer consists the opaque disc which is mounted on the shaft whose speed needs to be measured. The disc consists the equivalent holes around the periphery.



When the opaque portion comes in the line of light source and sensor, then the disc blocks the light source, and the output becomes zero. The production of pulses depends on the following factors:-

Advantages of Photoelectric Tachometer -

1. The digital output voltage is obtained, and hence there is no need of analogue to digital conversion.
2. The Pulses of constant amplitude are obtained which simplify the electronic circuit.

Disadvantages of Photoelectric Tachometers

1. The life of the light source is approximately 50,000 hours. Hence the light source needs to be replaced timely.
2. The accuracy of this method depends on the error which is represented by the unit pulse. These error can be minimized by using the gating period.

Advantages of Toothed Rotor Variable Rel. Tachometer

- ① It is simple and rugged in construction.
- ② It is maintenance free.
- ③ It is easy to calibrate.

Toothed Rotor Variable Reluctance Tachometer

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This Tachometer generator consists of a metallic toothed rotor mounted on the shaft whose speed is to be measured. A magnetic pickup is placed near the toothed rotor.

The magnetic pickup consists of a housing containing a small permanent magnet with a coil wound around it. When the rotor rotates, the reluctance of the air-gap between pickup and the toothed rotor changes giving rise to an induced e.m.f. in the pickup coil. The output is in the form of pulses, with a variety of wave shapes.

The frequency of the pulses of induced voltage will depend upon the number of teeth of the rotor and its speed of rotation. Since the number of teeth is known, the speed of rotation can be determined by measuring the frequency of pulses with an electronic counter. Suppose the rotor has "T" teeth, the speed of rotation is "n" rps and number of pulses per second is "P".

Number of pulses Per revolution = T

Galaxy M 55

$$\text{Hence speed, } n = \frac{\text{Pulses per second}}{\text{Number of Teeth}} = \frac{P}{T} \text{ rps}$$
$$= \frac{P}{T} \times 60 \frac{\text{rpm}}{\text{rps}}$$

MAGNETIC PICK-UP SENSORS

A magnetic pick-up sensor is normally a self-contained unit, making it hard to guess what is inside.

- ① - Inside the sensor body there is a permanent magnet with an attached pole piece. There is also a wire coil connected to the sensor outputs.

Operation Inside a Sensor:-

- ① Magnetic pick-up sensors are most often used to detect the speed of a rotating ferrous target.
- ② The sensor must be placed close enough to the object being detected that the target object will pass through the sensor's magnetic field.
- ③ A ferrous object passing through the magnetic field changes the shape of the field. In this example, each gear tooth is ferrous, so each tooth disrupts the field.
- ④ And because the magnetic field is drawn to the gear tooth it distorts slightly back and forth.

3) The magnetic field is also affected by the size of the change in the air gap between the sensor and the object being sensed.

4) A large change in the size of the air gap will have a greater effect on the magnetic field than a small change.

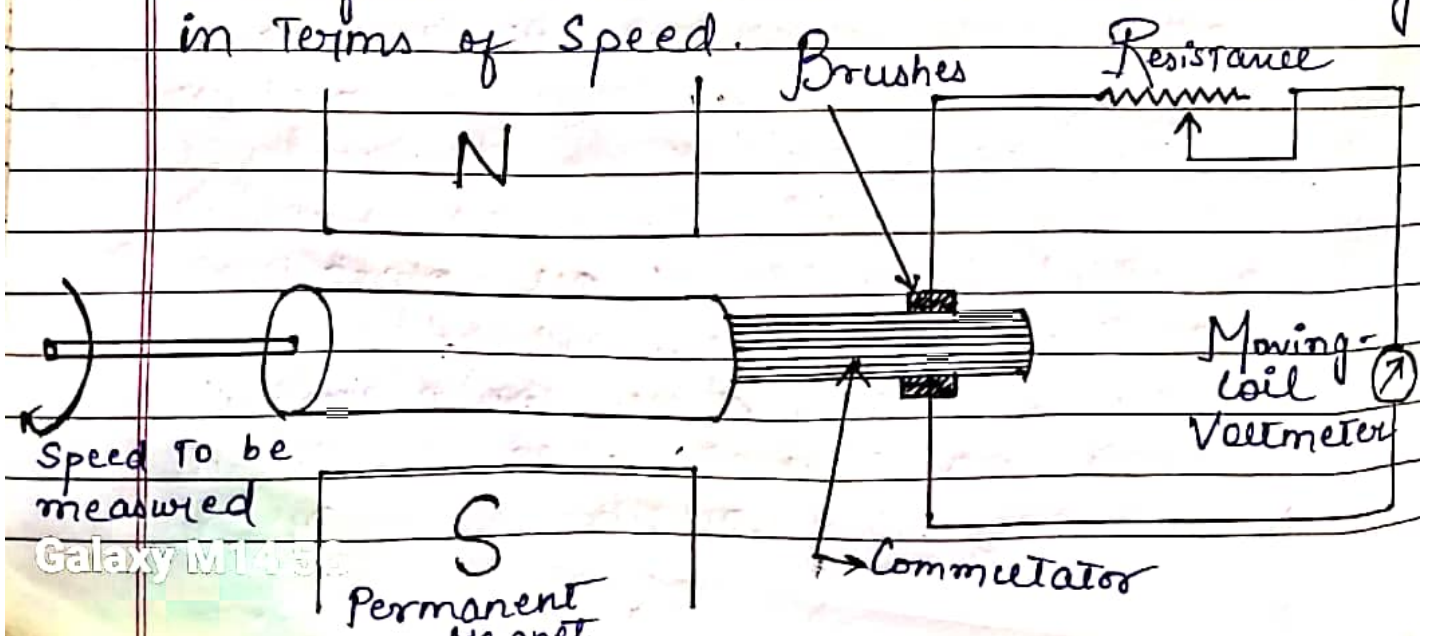
5) All this changing in the shape of the magnetic field induces a voltage in the coil.

Electromagnetic Tachometer Generators
There are two types of Electromagnetic Tachometer generators called Tachogenerators. These are -

① D.C Tachometer Generators

② A.C Tachometer Generators

① D.C Tachometer Generators - D.C Tachometer generator consists of a small armature which is coupled to the machine whose speed is to be measured. This armature revolves in the field of a Permanent magnet. The Emf generated is proportional to the product of flux and speed. The Polarity of output voltage indicates the direction of Rotation. This emf is measured with the help of a moving coil voltmeter having a Uniform scale and calibrated directly in terms of Speed.



Advantages :-

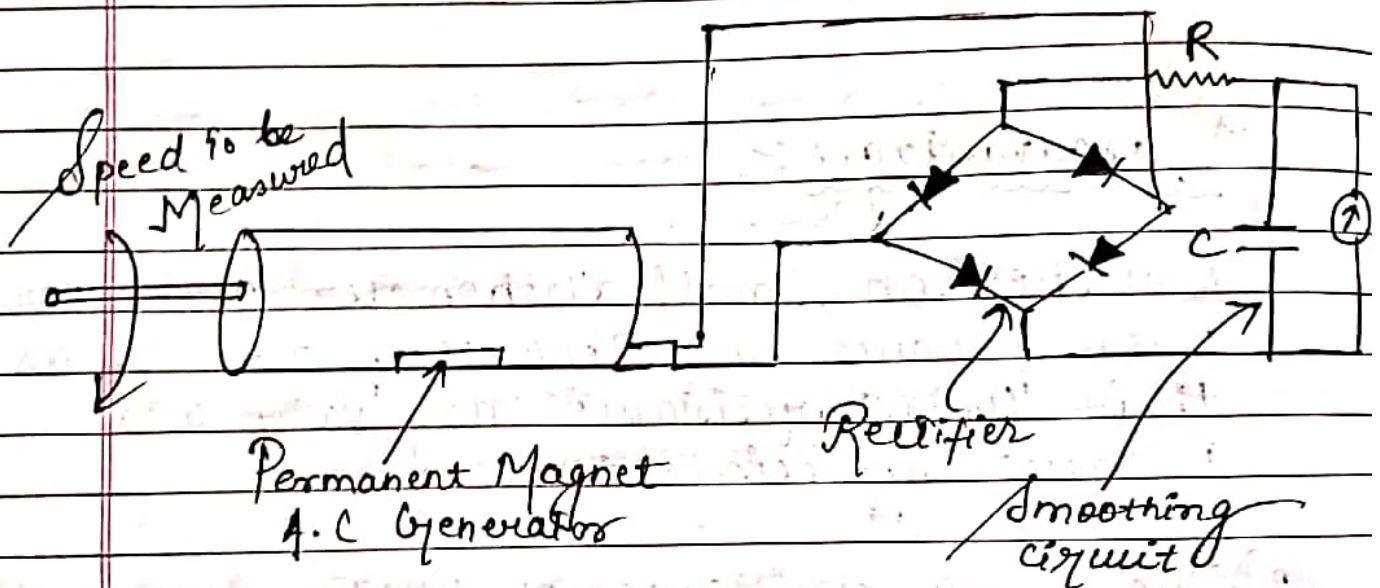
- ① The direction of rotation is directly indicated by the Polarity of the output voltage.
- ② The output voltage is typically 10 mV/rpm and can be measured with conventional type d.c. Voltmeter.

Disadvantages :-

- ① Brushes on small Tachometer generators often produce maintenance problems, as their contact resistance may vary and produce appreciable error.
- ② The input resistance of Meter should be very high as compared with output Resistance of Generator.

② A.C Tachometer Generator - In order to overcome some of the difficulties mentioned above, a.c Tachometer generators are used. The Tachometer Generator has rotating magnet which may be either a permanent magnet or an electromagnet. The coil is wound on the stator and therefore the problems associated with Commutator (as in d.c Tachometers) are absent.

The rotation of the magnet causes an emf to be induced in the rotor coil. The amplitude and frequency of this emf are both proportional to the speed of rotation. Thus either Amplitude or frequency of induced voltage may be used as a measure of Rotational Speed.



When Amplitude of induced voltage is used as a measure of speed, the output voltage of a.c. tachometer generator is rectified and is measured with a Permanent Magnet using moving coil instrument.

Advantages -

- ① The drag cup type of a.c. Tachogenerator described above is in fact a Two phase Squirrel Cage Induction Motor. One of the windings is excited by Carrier and the other windings.

② A linear relationship between the output voltage and speed is obtained if the carrier frequency is 5 to 10 times larger than the frequency of the speed signal.

③ The drag Cup Type Tachometers are rugged and wear-in - expensive.

④ In some situations these Tachometers are very useful since they give a ripple free output.

Disadvantages:-

① The output voltage is Proportional To the Product of speed and input voltage.

② These Tachometers are hard to Calibrate.

Toothed Rotor Variable Reluctance Tachometer

The tachometer generator consists of a metallic toothed rotor mounted on the shaft whose speed is to be measured. This is shown in fig - 29.47.

A magnetic pickup is placed near the toothed rotor.

Q A Variable Reluctance Type Tachometer has 60 rotor teeth. The counter record 3600 counts per second. Determine the speed in rpm.

Sol.
$$\text{Speed} = \frac{\text{Pulses Per Second}}{\text{Number of Teeth}} = \frac{3600 \text{ rps}}{60}$$

$$= \frac{3600 \times 60 \text{ rpm}}{60}$$

$$\boxed{\text{Speed} \Rightarrow 3600 \text{ rpm}}$$

Q An inductive pickoff operating from a 120 tooth wheel is used with a digital frequency meter to measure the speed of rotation of the shaft on which the wheel is mounted. The gating period is set to $10^4 \mu\text{s}$, and a reading of 0030 is obtained on the four digit display. What shaft speed does this represent in r.p.s?

If the available gating periods are $10^2, 10^3, 10^4, 10^5, 10^6, 10^7 \mu\text{s}$ respectively. What would be the optimum setting of gating period for making the measurement?

Q.1 ~~Q.1~~ ~~1059~~ Gating Period = $10^4 \mu s$

$$= 10^4 \times 10^{-6} s$$

Number of Pulses Per Second = $\frac{\text{Reading of digital meter}}{\text{gating Period}}$

$$= \frac{0030}{10^4 \times 10^{-6}}$$

$$= 3000$$

Speed = $\frac{\text{Number of Pulses Per Second}}{\text{Number of Teeth}}$

$$= \frac{3000}{120} \Rightarrow 25 \text{ rps}$$

$$= 25 \times 60 \text{ rpm}$$
$$= 1500 \text{ rpm}$$

Stroboscope and Stroboscopic Method

The stroboscope is a simple, portable manually operated device which may be used for measurement of periodic and rotary motion. Basically, the instrument is a source of variable frequency flashing brilliant light, the flashing frequency being set by the operator.

The circuit used is based upon variable frequency oscillator which controls the flashing frequency. The speed is measured by adjusting the frequency so that the moving objects are visible only at specific interval of time.

The method of use of the stroboscope depends upon imperfect dynamic response of the human eye. If a strong light is caused to flash on a moving object which at the time each flash occurs, occupies a given position, the object will appear to be stationary. Therefore the method is useful for only those types of motion which occur regularly after a fixed interval of time, such as oscillation or rotation.

The stroboscope consists of a source of flashing light whose frequency can be varied and controlled. This source is called a strobometer.

StroboTron - It is hot Cathode gaseous discharge tube as shown in fig. It has a Cathode, an anode and two grids, one is called the inner grid and other as the outer grid.

If the Potential of the outer grid is increased or that of inner grid is ~~increased~~ ^{decreased} beyond a certain limit, the conduction starts. Once the conduction starts, it can be stopped only by removing the anode potential.

The Flashing of light is started by a signal from an oscillator or a multivibrator. When the tube is ionized and starts giving out flashes, the capacitor C is discharged and a heavy current drawn by it reduces the anode potential due to large voltage drop in Resistance R_1 . The ionization stops and so does the flashing of light. In the mean time the capacitor is recharged and waits for the next pulse to start the flashing again. The tube has a maximum flashing rate of 300 per second with effective discharge time of 10 and 100 μs . The discharge current may be as high as 300 A.

Shaft Speed Measurement

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A Distinctive Mark is made on the shaft or on a disc attached to the shaft as shown in fig.

A Stroboscope is made to flash light directly on the mark. The Flashing frequency is adjusted until the mark appears stationary. Under these conditions, the speed is equal to the flashing frequency provided that the approximate speed of the shaft is known in advance and the flashing frequency is not allowed to depart too much away from this value.

The scale of the stroboscope is calibrated in terms of speed which can be directly read off.

If the two conditions outlined above are not met or if there are several identical marks on the shaft like spokes of a wheel or jaws of a chuck, serious errors in the measurement may arise.

Consider first the case where there is only one mark as shown in fig. If the mark is at A every time a flash occurs, a stationary image appears. This is the case when $n = f$ where n = Speed of shaft, rps and f = flashing frequency, Hz.

However, if $n = 2f, 3f, 4f, \dots$ etc. i.e. and if Speed of shaft is any whole multiple of flashing frequency the mark would appear as stationary. Therefore, the mere fact that a stationary image is obtained does not guarantee the shaft speed with certainty.

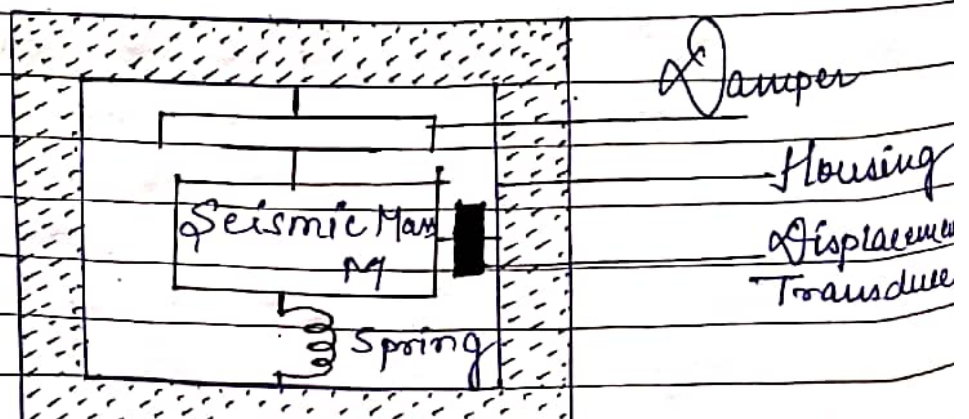
It should be noted that this Problem does not arise if $n < f$. Under these Circumstances if n is a sub-multiple of f , Multiple Images are obtained as shown in figs. -

Seismic Transducer

- * The Seismic Transducer is used for measuring the vibration of the ground. The spring mass damper element and the displacement transducer are the two main components of the seismic transducer.
- * The mass that is connected to the damper element and spring without any other support is known as spring mass damper element.
- * The seismic transducer is used for measuring the earth vibration, volcanic eruption and other vibrations etc.

Construction of Seismic Transducer

The systematic diagram of the seismic transducer is shown in the figure below. The mass is connected by the help of the damper and spring to the housing. The housing frame is connected to the source whose vibrations need to be measured.



The arrangement is kept in such a way so that the position of the mass remains same in the space. Such type of arrangement is kept for causing the relative motion between the housing frame and the mass.

Mode of Transducer-

The seismic transducer works in two different modes-

- Displacement Mode.
- Acceleration Mode.

The selection of the mode depends on the combinations of the mass, spring and damper combinations. The large mass and soft spring are used for the displacement mode measurement while the combinations of the small mass and stiff spring is used for the acceleration mode.

Types of Seismic Transducer-

Vibrometer and the accelerometer are the two types of the seismic transducer-

1. Vibrometer → The vibrometer or low-frequency meter is used for measuring the displacement of the body. It also measures the high frequency of the vibrating body.

2. Accelerometer $\rightarrow$ The accelerometer measures the acceleration of the measuring body. The acceleration shows the total force acting on the object.

$\rightarrow$ Accelerometers measure acceleration and detect changes in orientation or movement.

$\rightarrow$ Types of Accelerometer are Piezoelectric, Capacitive, and Piezoresistive accelerometers.

Types of Accelerometer

1) Potentiometric Type Accelerometer -

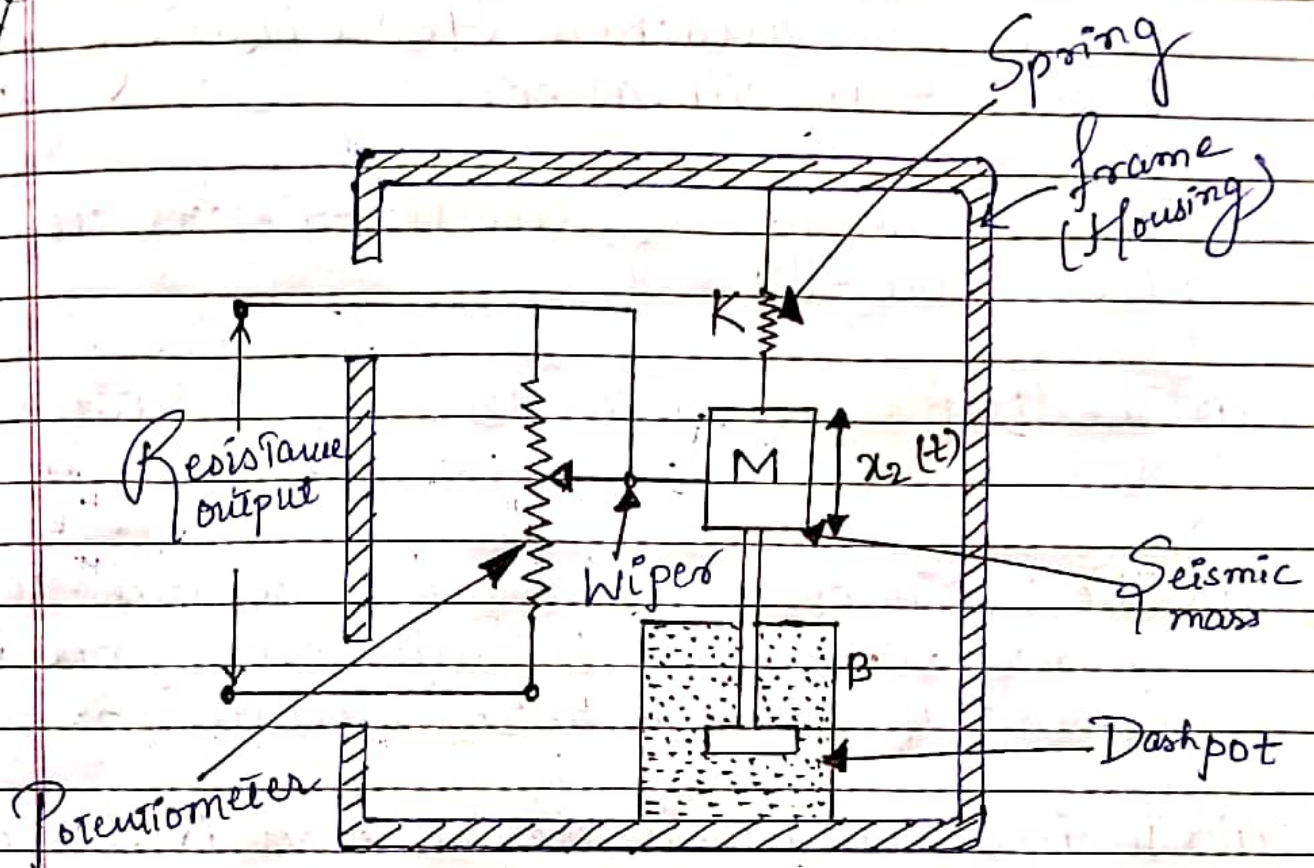
$\rightarrow$ This is the most simple type of Accelerometer as shown. The seismic mass is attached to the wiper arm or resistance potentiometer.

$\rightarrow$ The relative motion of the mass with respect to the transducer frame is sensed either as a change in resistance or as a change in voltage output.

$\rightarrow$ The damping may be provided by filling the housing of the accelerometer completely with a viscous fluid or it may be provided by a dashpot.

- The major drawbacks of a seismic accelerometer using resistance potential divider are its limited resolution and a rather low natural frequency.

Fig -

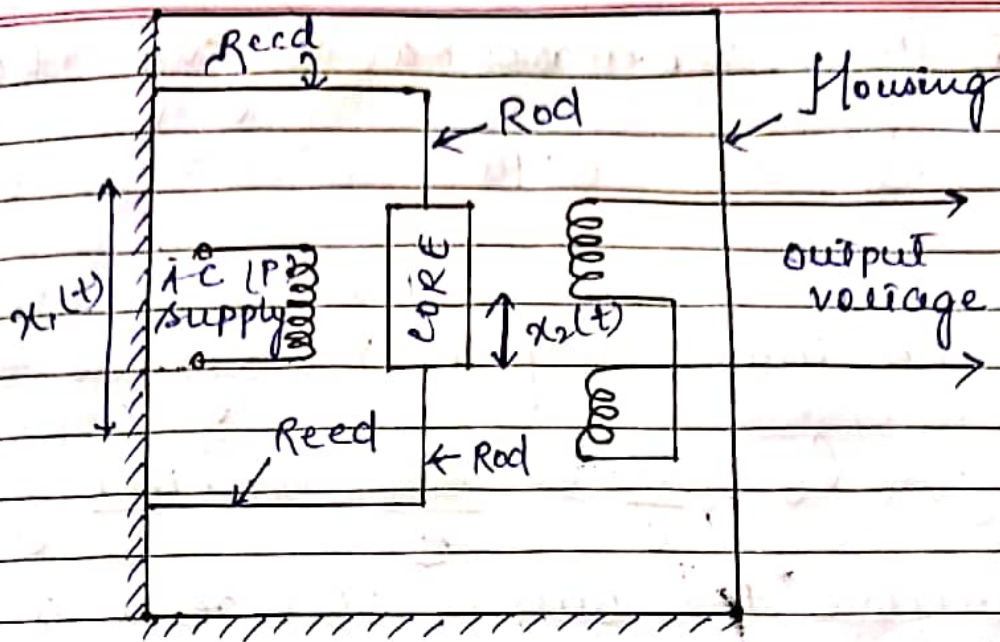


2) LVDT Accelerometer - LVDT Accelerometer shows a seismic accelerometer using a linear voltage differential transformer (LVDT). The core of the LVDT acts as the mass and two flexible reeds, attached at each end of the rods of the core, provide the necessary spring action. The reeds are attached to a housing which is subject to vibrations.

The advantages of accelerometers using LVDT are -

- (i) The LVDT has a much smaller mass and thus has a higher natural frequency.
- (ii) The LVDT offers a lower resistance to the motion than the potentiometer and is capable of a much better resolution.
- (iii) It is a contactless device and is therefore free from problems which arise on account of moving contacts.

The LVDT accelerometers are used for steady state and low frequency vibration measurement.



Piezo - Electric Accelerometers

Piezo - Electric Transducers have been explained in Details -

- * When a force F is applied to a Piezo-electric crystal it develops a charge $Q = dF$ coulomb, where d = charge sensitivity of crystal c/n.

Through incorporation of mass m , in direct contact with the crystal, we have the essential components of an Acceleration Transducer. By Applying a Varying acceleration to the mass - crystal assembly, the crystal experience a Varying force. The force is given by $F = m \times a$, where a = acceleration.

This force generates a Varying Charge

$$Q = dF = dMa$$

Suppose the crystal has a Capacitance C , the no load output voltage is -

$$V_o = \frac{Q}{C} = \frac{dF}{C} = \frac{dMa}{C}$$

Therefore the output voltage is a measure of the acceleration.

* The Piezo-electric Crystal is Spring loaded with Seismic mass in contact with the crystal. When subjected to an acceleration, the seismic mass stresses the crystal to a force $F = Ma$, resulting in a voltage generated across the crystal. This force, generates an output voltage which is proportional to the acceleration.

Some of the features of Piezo-electric accelerometers are -

① The instrument is quite small in size and has a small weight (Typically 0.025 kg).

② The natural frequency is very high. It may be as high as 100 kHz and the accelerometer is useful for

high frequency applications. They can be used for any vibration and shock applications.

(3) The crystal is a source with a high output impedance and in order to avoid loading effect, a voltage monitoring source of a high input impedance should be used.

(4) These accelerometers are useful for high input frequencies and their response is poor at low frequency.

Q An accelerometer has a seismic mass of 0.05 kg and a spring constant of 3×10^3 N/m. Maximum mass displacement is ± 0.02 m (before the mass hits the stop). Calculate (a) maximum measurable acceleration and (b) natural frequency.

Sol. Natural frequency $\omega_n = \sqrt{k/m}$

$$= \sqrt{\frac{3 \times 10^3}{0.05}} = 245 \text{ rad/s}$$

Maximum acceleration $a_m = \omega_n^2 x_m$

$$= (245)^2 \times (0.02)$$

$$= 1200 \text{ m/s}^2$$

(b) Natural frequency = $f = \frac{245}{2\pi} = 29 \text{ Hz}$